

FOREST ENCROACHMENT CHAINSAW (FECS)

Smart Chainsaw Monitoring System for
Reducing Deforestation



Prepared By

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Certification

This project report entitled **Forest Encroachment Chainsaw (FECS)** was developed by Frempong Boateng Prince as an innovative solution aimed at reducing illegal logging and improving environmental monitoring through smart technology integration.

Dedication

This project is dedicated to all individuals and organizations working tirelessly towards environmental conservation of trees and forests for a sustainable development.

Acknowledgements

I express my sincere gratitude to my family members, teachers, mentors and supporters who contributed to the successful completion of this project.

Abstract

Title: Smart Conservation: Reimagining Forest Protection in Ghana Through the FECS Project Deforestation remains one of the greatest environmental challenges facing the world today, and Ghana is no exception. For decades, the nation's lush forest reserves, from the Atewa Range to the Kakum conservation areas, have faced severe degradation due to illegal logging and "galamsey" (illegal mining). Traditional methods of forest policing—relying on a limited number of forestry guards to patrol vast, dense terrains—often turn into a losing game of hide-and-seek. By the time authorities hear the distant buzz of a chainsaw, the trees are already down. To effectively safeguard these natural resources, a shift from reactive policing to proactive technology is essential. The Forest Encroachment Chainsaw (FECS) project offers an innovative solution, combining web technologies, GPS tracking, and hardware control systems into a smart monitoring tool built specifically for the Ghanaian context.

At its core, the FECS system redefines forest conservation by putting a digital lock on the primary tool of destruction: the chainsaw itself. In practice, the system requires chainsaw operators to authenticate their identities through a web platform before the equipment can even be turned on. For a licensed timber worker in the Western Region, this means logging into a mobile-friendly portal before pulling the starter cord. If an unauthorized operator or illegal logger attempts to use a stolen or unregistered chainsaw, the hardware control system acts as a digital kill-switch, rendering the tool completely useless. This authentication phase shifts the power dynamic, stopping illegal logging before the first cut is made.

Once an operator is authenticated and the chainsaw is running, the system utilizes real-time GPS tracking to maintain strict oversight. The device actively pings its precise location back to a centralized dashboard managed by environmental authorities, such as the Forestry Commission in Accra or Kumasi. If a logger drifts outside their designated concession and enters a protected national park or a sacred grove, an automated alarm alerts officials instantly. Instead of wandering blindly through thousands of hectares of forest, authorities are given exact coordinates, allowing for swift, targeted interventions.

What makes the FECS project particularly viable for Ghana is its practical design, which accounts for the harsh realities of local terrain. Deep within tropical rainforests, there is virtually no access to grid electricity, and cellular network coverage can be highly

unstable. Acknowledging these hurdles, the prototype was developed using locally available materials and relies on renewable energy sources, such as compact, solar-charged batteries integrated directly into the housing. This ensures that the device remains operational in remote areas without relying on constant battery replacements or external power sources, making it a rugged and sustainable tool for long-term field deployment.

Ultimately, the FECS project serves as a powerful demonstration of how the Internet of Things (IoT) can be applied to environmental protection and forest conservation. By transforming a standard, mechanical tool into a connected, "smart" device, the system bridges the gap between hardware engineering and digital policy enforcement. For Ghana, adopting such IoT-driven solutions offers a path forward to protect its rich biodiversity, secure timber revenues, and preserve its natural heritage for future generations.

Contents

Certification	i
Dedication	ii
Acknowledgements	iii
Abstract	iv
List of Figures	viii
List of Tables	ix
1 Introduction	1
1.1 Background	1
1.2 Problem Statement	1
1.3 Purpose of the Project	1
2 Project Objectives	2
2.1 Main Objective	2
2.2 Specific Objectives	2
3 Concept and Design	3
3.1 Overview	3
3.2 System Architecture	4
4 Web-Based Component	5
4.1 ESP-32 Development Board	5
4.2 Authentication System	5
4.3 GPS Monitoring	6
5 Hardware Component	7
5.1 Prototype Materials	7
5.2 Electronic Components	7
5.3 Power Supply	8

List of Figures

1	FECS Symbolic Electrical Schematic Diagram	19
2	FECS Physical Interface Hardware Connection Layout	20

List of Tables

1

Introduction

1.1 Background

Deforestation is a major environmental problem that contributes to climate change, biodiversity loss and ecosystem degradation.

1.2 Problem Statement

Illegal logging activities continue to threaten forest resources due to inadequate monitoring and weak identification systems. Due to high demand for wood, unauthorised lumberjacking has been the only means for some lumberjacks to obtain more tree woods to sell to fill their pockets. This practice has gradually and consequently reduced the forest mass in the country and in turn reduces the circulation of carbon dioxide in the atmosphere, and this plays an overall factor in global warming.

1.3 Purpose of the Project

The FECS project seeks to introduce a smart chainsaw system capable of tracking and monitoring logging activities by competent bodies.

3

Concept and Design

3.1 Overview

The Forest Encroachment Chainsaw is a web + hardware chainsaw that has been made to reduce deforestation and to monitor the actions of the lumberjack. The concept behind its development is simple but needed hard work to bring this idea to life.

The FECS is in two folds, the web-based aspect and the hardware aspect. The web-based aspect it built on the ESP-32 + Wi Fi development board. The lumberjack connects to the development board through a Wi Fi connection. Once the connection is secured, the lumberjack logs in to the website, “FECS IDENTIFICATION GHANA”. Here, he will be made to fill various forms such as Chainsaw ID, Security Code and EPA number (standing for Environmental Protection Agency) in order to secure his ownership on the chainsaw, to be able to operate the chainsaw and to be tracked by the EPA. Once he has logged in correctly, his/her Global Positioning System (GPS) is on, to prompt the EPA that his or her chainsaw is in use. This enables the EPA to monitor the activities of the lumberjack and to also know who to question if there is an unauthorized cutting down of trees.

Due to our restrictions as students, we made a prototype out of local materials. For the hardware, local materials such as Cardboard, Ply Wood, Sticking Glue, Chopsticks and paper sheets were included. The ESP-32 development board is placed in the chainsaw, powered by a 9V Hiwatt battery. Other electronic devices such as the Yellow Electric Motor 20000rpm, Jumper wires, a motor driver, a relay module and a switch were all included in the project. These devices are powered with solar energy (green energy). As the development board is connected to the electric motor, the lumberjack logs in into his FECS account and this triggers the motor to run, moving the chain around the chain rim.

The idea me and my team members are trying to put in action, is being able to integrate a web-based hardware device to curb deforestation which has now been identified

to be the pathway to destruction for our world. With all these equipment add-ons, FECS is still targeted to be less cheap than the normal chainsaw if the project is held. I believe I and my partners of the Tectitans STEM team have done a fulfilling job.

3.2 System Architecture

Insert system architecture diagram here.

4

Web-Based Component

4.1 ESP-32 Development Board

The ESP-32 serves as the communication bridge between the hardware and web platform.



4.2 Authentication System

FECS IDENTIFICATION GHANA

EPN Number:

Chainsaw ID:

Security Code:

Submit

© 2024 FECS IDENTIFICATION GHANA

Users provide:

- Chainsaw ID
- Security Code

- EPA Number

4.3 GPS Monitoring

GPS enables real-time monitoring of chainsaw activities of the lumberjack by the Environmental Protection Agency and the Forestry Commission.

5

Hardware Component

5.1 Prototype Materials

- Cardboard
- Plywood
- Glue
- Chopsticks
- Paper Sheets

5.2 Electronic Components

- ESP-32 Development Board
- 9V Battery
- Motor Driver
- Relay Module
- Electric Motor
- Jumper Wires
- Switch
- GPS Tracker

5.3 Power Supply

Solar energy is used to power the system. This is to minimize electricity usage since chainsaws operate on high voltages to provide enough torque for the motor to drive the chainsaw teeth at high speeds. The solar energy is harnessed by solar panels implanted in the device.

6

Operational Mechanism

6.1 System Workflow

1. The user connects to ESP-32 via Wi-Fi on his or her device, whether a phone or a laptop. This will not require internet connection.
2. The user then gains access the FECS website. The website is offline and can be accessed without internet connection only if the device of the user is connected to the ESP-32 in his or her Chainsaw.
3. The User logs in with his or her EPA number, Chainsaw ID and Security Code. The user's details are unique to his or her chainsaw and can only function with the chainsaw close to him or her since the ESP-32 uses Wi-Fi.
4. The GPS tracker is activated to provide the Environmental Protection Agency and the Forestry Commission location of live users and their registered details. This is done to easily monitor and locate unauthorised lumberjacking.
5. Motor receives signal. This signal is only received when the log in details of the lumberjack is accurate and valid. The relay Module connected to the Motor operates on the NAND logic gate function, where all the inputs of the lumberjack must be correct to enable the chainsaw to function.
6. The Chainsaw operates. In case the lumberjack decides to destroy the integrated system of the ESP-32 and the Relay Module, he destroys the chainsaw alongside and renders the chainsaw functionally useless. The EPA or the Forestry Commission can decide on the time duration of use of the chainsaw of which if elapsed the chainsaw automatically goes off.

7

Environmental Significance

7.1 Reduction of Illegal Logging

The system discourages unauthorized logging.

7.2 Forest Conservation

The project contributes to the preservation of forests.

7.3 Climate Change Mitigation

Reduced deforestation supports proper cycling of the carbon and oxygen cycle and carbon sequestration.

8

Innovation and Uniqueness

The FECS project integrates IoT, GPS tracking, authentication systems and renewable energy into a single environmental protection device.

9

Challenges Encountered

- Limited budget.
- Limited access to advanced equipment.
- Constraints in prototype construction.
- Student resource limitations.

Future Improvements

Future developments may include

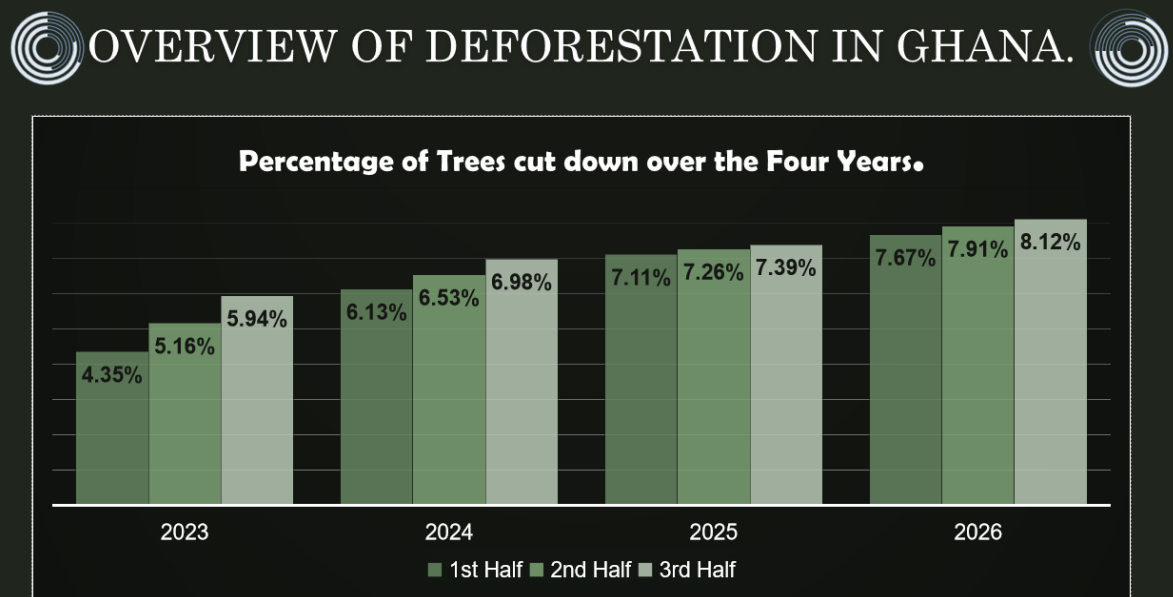
- Integration of mobile applications to make it accessible to all.
- Cloud-based monitoring and decentralization of the monitoring accessibility to the local Environmental Protection bodies in the communities and villages to reduce the burden of the management of the whole system across the country.
- Biometric authentication.
- Enhanced GPS accuracy for more precise locations.
- Industrial-grade construction.

Conclusion

The Forest Encroachment Chainsaw (FECS) project demonstrates how technology can be leveraged to combat illegal logging and support environmental sustainability. Through the integration of IoT, GPS tracking and web-based authentication systems, the project provides a promising solution for forest monitoring and conservation.

References

This Project is referenced from Yearly-based statistics for the past four years of Deforestation in Ghana.



From the Chart, it can be observed that, from the first 4 months of 2021 through to the last four months of 2024, there is a gradual increase in the percentage of cut down trees from 4.35%. Also, it can be observed that, for each year, there is a slight rise in the percentage of the cut down trees. Though the Forestry commission is doing well with the Green Ghana initiative, it is not enough to curb the menace of deforestation.

1. Environmental Protection Agency (EPA) Reports
2. Forestry Commission Publications
3. Internet of Things (IoT) Literature
4. Renewable Energy Resources

Images of some parts of Prototype

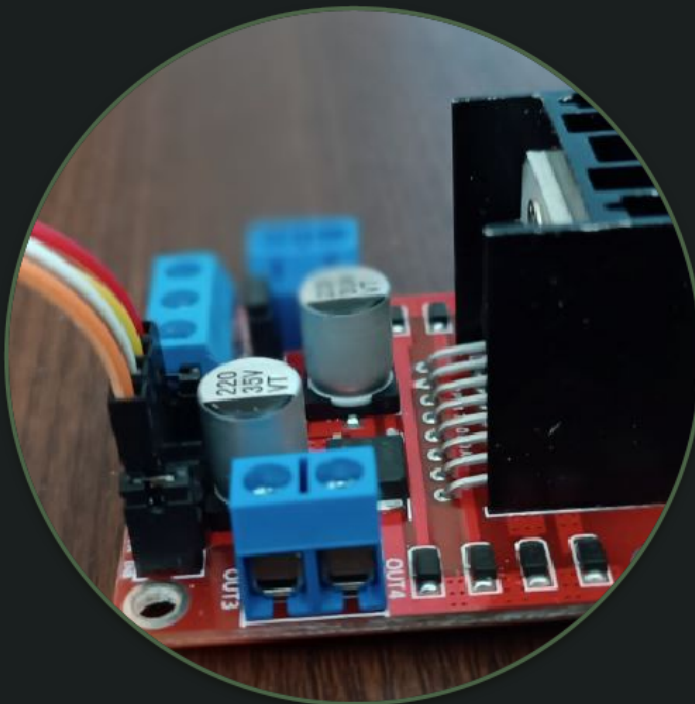
- ESP-32 + Wi-Fi



- Relay Module



- Motor Driver



- Website

F ECS IDENTIFICATION GHANA

EPA Number:

Chainsaw ID:

Security Code:

Submit

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Circuit Diagrams

This appendix provides a comprehensive look at the electrical design of the Forest Encroachment Chainsaw (FECS) system through two distinct structural layouts.

.1 System Schematic (Symbolic Logic Diagram)

The diagram below utilizes standard electronic engineering symbols to trace the logic circuit, signal routing, and power distribution paths from the solar panel and batteries through the microcontroller unit to the actuation hardware.

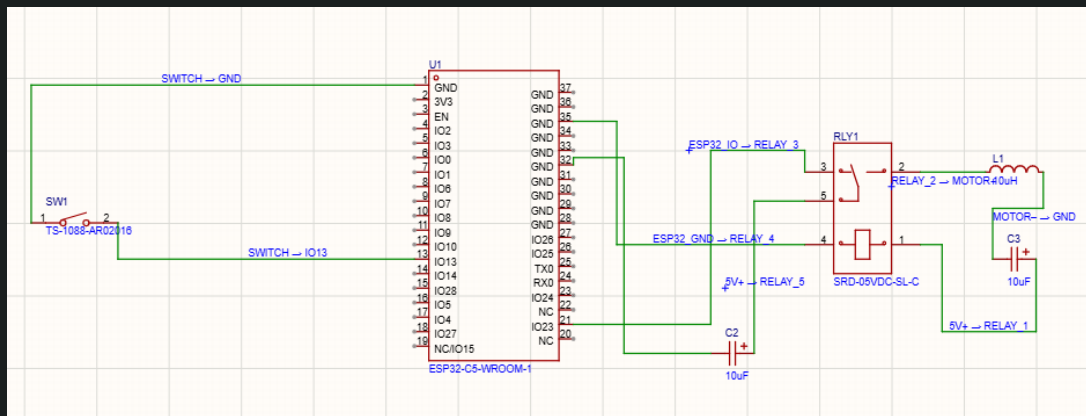


Figure 1: FECS Symbolic Electrical Schematic Diagram

.2 Physical Wiring and Connection Layout

The diagram below provides a physical component layout map, depicting the exact GPIO pins of the ESP-32 platform, terminal block interfaces of the relay mechanism, and wiring connections used during prototype assembly.

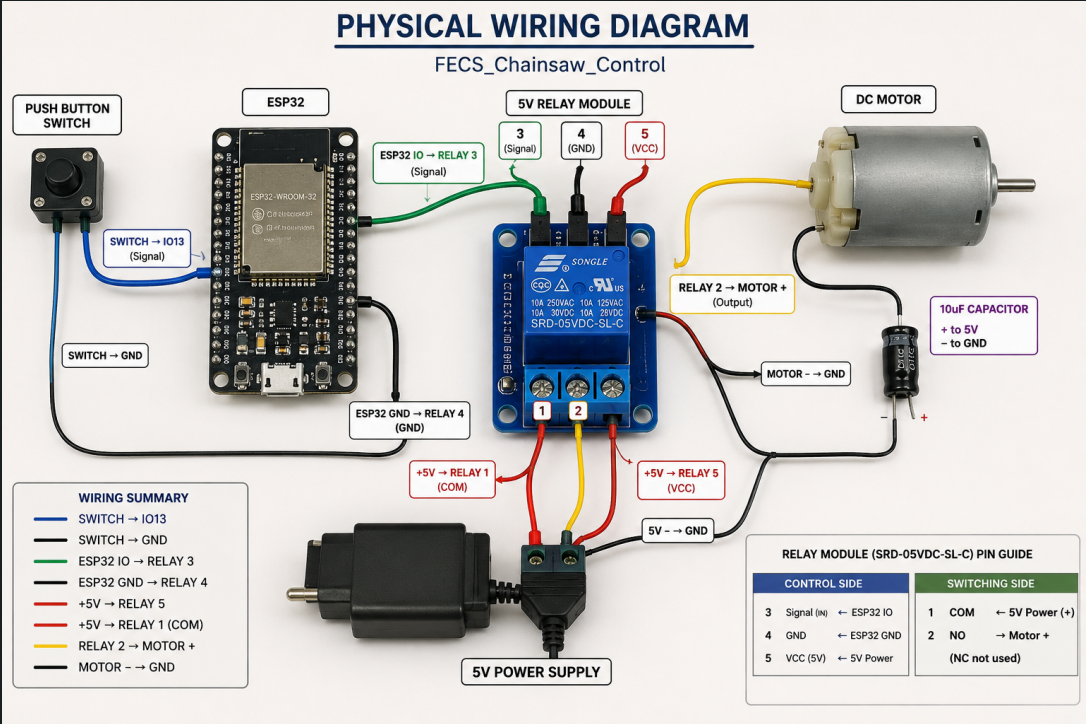


Figure 2: FECS Physical Interface Hardware Connection Layout

Source Code

2.1 ESP32 Access Point & Motor Control (Arduino C++)

The firmware below configures the micro-controller as an Access Point (AP), Wi-Fi logs serving the local portal login page, handling authentication rules, and controlling the driver relay logic pin.

```
1 #include <WiFi.h>
2 #include <WebServer.h>
3
4 // --- Configuration ---
5 // The Wi-Fi credentials for the ESP32 Access Point
6 const char* ssid = "FECS_Chainsaw_01";
7 const char* password = "fecsforachange123";
8
9 // The GPIO pin connected to your Relay Module (which controls the
   motor driver)
10 const int RELAY_PIN = 26;
11
12 // Mock Database Credentials (for prototype testing)
13 const String validEPA = "EPA123";
14 const String validID = "CS001";
15 const String validCode = "7788";
16
17 // Create a web server listening on port 80
18 WebServer server(80);
19
20 // --- HTML & CSS Code (Matches your uploaded image) ---
21 // We use a raw string literal R"rawliteral(...)rawliteral" to
   store the HTML easily
22 const char index_html[] PROGMEM = R"rawliteral(
23 <!DOCTYPE html>
24 <html lang="en">
25 <head>
26     <meta charset="UTF-8">
27     <meta name="viewport" content="width=device-width, initial-
   scale=1.0">
28     <title>FECS IDENTIFICATION</title>
29     <style>
30         body {
31             font-family: Arial, sans-serif;
32             background-color: #f4f7f9;
```



```

84         background-color: #0056b3;
85     }
86     .footer {
87         margin-top: 30px;
88         font-size: 10px;
89         color: #aaa;
90     }
91     .error {
92         color: red;
93         font-size: 12px;
94         text-align: center;
95         margin-bottom: 10px;
96         display: none;
97     }
98     </style>
99 </head>
100 <body>
101
102     <h2>FECS IDENTIFICATION GHANA</h2>
103
104     <div class="card">
105         <div class="error" id="errorMsg">Invalid Credentials. Try
106         Again.</div>
107         <form action="/login" method="POST">
108             <div class="form-group">
109                 <label>EPA Number:</label>
110                 <input type="text" name="epa" placeholder="Enter
111                 EPA Number" required>
112             </div>
113
114             <div class="form-group">
115                 <label>Chainsaw ID:</label>
116                 <input type="text" name="id" placeholder="Enter
117                 Chainsaw ID" required>
118             </div>
119
120             <div class="form-group">
121                 <label>Security Code:</label>
122                 <input type="password" name="code" placeholder="
123                 Enter Security Code" required>
124             </div>
125
126             <button type="submit">Submit</button>
127         </form>
128     </div>
129
130     <div class="footer">&copy; 2024 FECS IDENTIFICATION GHANA</div>
131 </body>
132 </html>

```

```

129         // Check URL parameters to show error message if login
        failed
130         const urlParams = new URLSearchParams(window.location.
        search);
131         if(urlParams.has('error')) {
132             document.getElementById('errorMsg').style.display = '
        block';
133         }
134     </script>
135 </body>
136 </html>
137 )rawliteral";
138
139 // --- HTML Success Page ---
140 const char success_html[] PROGMEM = R"rawliteral(
141 <!DOCTYPE html>
142 <html>
143 <head>
144     <meta name="viewport" content="width=device-width, initial-
        scale=1.0">
145     <style>
146         body { font-family: Arial; text-align: center; padding-top
        : 50px; background-color: #e8f5e9;}
147         h1 { color: #2e7d32; }
148     </style>
149 </head>
150 <body>
151     <h1>Authentication Successful!</h1>
152     <p>GPS Tracker Activated.</p>
153     <p>Motor Unlocked. You may now operate the chainsaw.</p>
154 </body>
155 </html>
156 )rawliteral";
157
158
159 // --- Server Handlers ---
160
161 void handleRoot() {
162     server.send(200, "text/html", index_html);
163 }
164
165 void handleLogin() {
166     // Check if form data was sent
167     if (server.hasArg("epa") && server.hasArg("id") && server.hasArg
        ("code")) {
168         String inputEPA = server.arg("epa");
169         String inputID = server.arg("id");
170         String inputCode = server.arg("code");
171
172         // Verify credentials against our mock database

```

```
173     if (inputEPA == validEPA && inputID == validID && inputCode ==
174         validCode) {
175         Serial.println("Login Successful! Turning on motor and GPS
176         ...");
177
178         // TRIGGER THE RELAY / MOTOR DRIVER
179         digitalWrite(RELAY_PIN, HIGH);
180
181         // Send success page to user
182         server.send(200, "text/html", success_html);
183     } else {
184         Serial.println("Login Failed.");
185         // Redirect back to root with an error flag
186         server.setHeader("Location", "/?error=1");
187         server.send(303);
188     }
189 } else {
190     server.send(400, "text/plain", "Bad Request");
191 }
192
193 // --- Setup and Loop ---
194
195 void setup() {
196     Serial.begin(115200);
197
198     // Configure Relay Pin
199     pinMode(RELAY_PIN, OUTPUT);
200     digitalWrite(RELAY_PIN, LOW); // Ensure motor is OFF by default
201
202     // Setup ESP32 as Access Point
203     Serial.println("Setting up Access Point...");
204     WiFi.softAP(ssid, password);
205
206     IPAddress IP = WiFi.softAPIP();
207     Serial.print("AP IP address: ");
208     Serial.println(IP);
209
210     // Define routing
211     server.on("/", HTTP_GET, handleRoot);
212     server.on("/login", HTTP_POST, handleLogin);
213
214     // Start the server
215     server.begin();
216     Serial.println("Web server started.");
217 }
218
219 void loop() {
220     // Listen for incoming clients
221     server.handleClient();
222 }
```

.2.2 Portal Interface Frontend (HTML & CSS)

The implementation block below outlines the structure of the authentication portal served to operators by the microcontroller system.

```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4     <meta charset="UTF-8">
5     <meta name="viewport" content="width=device-width, initial-
6     scale=1.0">
7     <title>FECS IDENTIFICATION</title>
8     <style>
9         /* This matches the greyish-blue background from your
10        screenshot */
11        body {
12            font-family: Arial, sans-serif;
13            background-color: #f4f7f9;
14            display: flex;
15            flex-direction: column;
16            align-items: center;
17            justify-content: center;
18            height: 100vh;
19            margin: 0;
20        }
21
22        /* The bright blue heading */
23        h2 {
24            color: #007bff;
25            font-size: 24px;
26            letter-spacing: 0.5px;
27            margin-bottom: 20px;
28            text-transform: uppercase;
29        }
30
31        /* The white floating card box container */
32        .card {
33            background-color: white;
34            padding: 35px;
35            border-radius: 8px;
36            box-shadow: 0 4px 15px rgba(0, 0, 0, 0.05);
37            width: 320px;
38        }
39
40        .form-group {
41            margin-bottom: 20px;
42        }
43
44        /* Small, bold labels right above the text inputs */
45        label {
46            display: block;
47            font-size: 13px;

```

```
46         font-weight: bold;
47         color: #333;
48         margin-bottom: 6px;
49     }
50
51     /* The input fields styling */
52     input[type="text"], input[type="password"] {
53         width: 100%;
54         padding: 12px;
55         border: 1px solid #ccc;
56         border-radius: 4px;
57         box-sizing: border-box;
58         font-size: 14px;
59         background-color: #fff;
60     }
61
62     /* Subtle gray placeholder text styling */
63     input::placeholder {
64         color: #aaa;
65     }
66
67     /* The solid blue Submit button */
68     button {
69         width: 100%;
70         padding: 12px;
71         background-color: #007bff;
72         color: white;
73         border: none;
74         border-radius: 4px;
75         font-size: 15px;
76         font-weight: bold;
77         cursor: pointer;
78         margin-top: 10px;
79         transition: background 0.2s;
80     }
81
82     button:hover {
83         background-color: #0056b3;
84     }
85
86     /* Tiny grey copyright footer at the very bottom */
87     .footer {
88         margin-top: 25px;
89         font-size: 11px;
90         color: #888;
91     }
92
93     /* Hidden error box that reveals if someone types the
wrong code */
94     .error {
95         color: #d9534f;
```

```

96         font-size: 13px;
97         text-align: center;
98         margin-bottom: 15px;
99         display: none;
100         font-weight: bold;
101     }
102 </style>
103 </head>
104 <body>
105
106     <h2>FECS IDENTIFICATION GHANA</h2>
107
108     <div class="card">
109         <div class="error" id="errorMsg">Invalid EPA, ID, or Code!
110     </div>
111
112         <form action="/login" method="POST">
113             <div class="form-group">
114                 <label>EPA Number:</label>
115                 <input type="text" name="epa" placeholder="Enter
116 EPA Number" required>
117             </div>
118
119             <div class="form-group">
120                 <label>Chainsaw ID:</label>
121                 <input type="text" name="id" placeholder="Enter
122 Chainsaw ID" required>
123             </div>
124
125             <div class="form-group">
126                 <label>Security Code:</label>
127                 <input type="password" name="code" placeholder="
128 Enter Security Code" required>
129             </div>
130
131             <button type="submit">Submit</button>
132         </form>
133     </div>
134
135     <div class="footer">&copy; 2024 FECS IDENTIFICATION GHANA</div>
136 >
137
138     <script>
139         // Simple logic: if the micro-controller tells us login
140         failed, show the error alert
141         const urlParams = new URLSearchParams(window.location.
142 search);
143         if (urlParams.has('error')) {
144             document.getElementById('errorMsg').style.display = '
145 block';
146         }
147     </script>

```

